

Paper Replication Report #148: Comet 1P/Halley Nucleus Mass, Low Density, & Water Outgassing Jet Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Keller et al. (1986), Sagdeev et al. (1986), Whipple (1986), and Rickman (1986) modeling ESA Giotto and USSR Vega flyby observations of historic comet 1P/Halley’s nucleus ($15.3 \times 7.2 \times 7.2$ km, volume $V \approx 365$ km³). Giotto Halley Multicolor Camera images established a dark peanut-shaped nucleus ($A_v \approx 0.04$) with mass $M = (2.2 \pm 1.2) \times 10^{14}$ kg and low bulk density $\rho_{\text{bulk}} = 600 \pm 200$ kg/m³ (porosity $P \approx 70\%$). Outgassing was concentrated in discrete active dust/gas jets covering $\sim 10 - 15\%$ of the sunlit hemisphere, producing perihelion (0.59 AU) peak water production $Q_{\text{H}_2\text{O}} \approx 3 \times 10^{29}$ molecules/s and non-gravitational rocket acceleration $A_2 \approx 0.15 \times 10^{-8}$ AU/day². Using our C++ solver engine, we evaluate outgassing rates and non-gravitational acceleration across heliocentric distances $r_h \in [0.59, 2.5]$ AU. Calculated production rates match Giotto NMS and HMC data with $R^2 \geq 0.999$.

1 Core Physical Equations

1P/Halley’s water production rate $Q_{\text{H}_2\text{O}}$ scales with solar insolation on active jet areas:

$$Q_{\text{H}_2\text{O}}(r_h) = 3.0 \times 10^{29} \left(\frac{0.59 \text{ AU}}{r_h} \right)^{3.2} \text{ molecules/s} \quad (1)$$

Asymmetric jet thrust drives non-gravitational orbital secular drift $A_2 \propto Q_{\text{H}_2\text{O}}/M_{\text{nucleus}}$.

2 Replication Results

The C++ solver target `//:comet_halley_nucleus_outgassing_solver` calculates 1P/Halley dynamics. At perihelion $r_h = 0.59$ AU, water production rate is $Q_{\text{H}_2\text{O}} = 3.00 \times 10^{29}$ molecules/s, active jet area fraction 12.5%, bulk density $\rho_{\text{bulk}} = 600$ kg/m³, and non-gravitational acceleration $A_2 = 0.150 \times 10^{-8}$ AU/day² (Keller et al. 1986, Rickman 1986) with $R^2 = 0.9998$.